15 Summary of Environmental Outcome

15.1 General

15.1.1.1 This chapter summarises the overall environmental outcomes due to the Yuen Long South (YLS) development, comprising a total area of 223.5ha. It can been seen from **Sections 1 to 3** that environmental considerations have been the key considerations throughout the development of proposed land uses as illustrated in the RODP. Alternative options for land uses, designs and construction methodologies have been duly considered throughout the process to response to the public aspirations collated during the public engagement exercises in an appropriate manner. Besides, all the options considered have ensured that environmental impacts could be avoided or minimised where practicable and mitigated by implementing suitable mitigation measures to fulfil all the statutory requirements. The technical assessments conducted (see **Section 4 to Section 13**) have demonstrated that all the statuary requirements in EIA Study Brief (EIA SB No.: ESB-246/2012) and Technical Memorandum on Environmental Impact Assessment (EIAO-TM) have been complied with.

15.1.1.2 The following sections summarise the approaches that have been adopted to either avoid or minimise various environmental impacts throughout the design process, and the associated environmental enhancements and compensation plans.

15.2 Environmentally Friendly Options Recommended to Avoid Environmental Impacts

15.2.1.1 Avoidance of environmental impacts has been one of the key considerations throughout the entire project development and design. For those impacts that could not be avoided, due considerations have been given to minimise those impacts as much as practicable so that all the residual impacts would comply with the statutory requirements. For YLS development in particular, given the views and opinions collated from public engagement exercises, much emphasis has been given to address the natural resources such as continuous watercourses, mature woodland, agricultural land near Shan Ha Tsuen, Tai Tong (Pak Sha Tsuen) Egretry, built heritages etc. All these approaches to avoid environmental impacts have been suitably incorporated into the RODP. A summary of these approaches is given below:

- Avoidance of recognised sites of conservation importance;
- Avoidance of direct impact on watercourses of higher ecological value;
- Avoidance of impacts to Tai Tong (Pak Sha Tsuen) Egretry;
- Avoidance of encroachment on existing villages; and
- Avoidance of encroachment on built heritages.

15.2.2 Avoidance of recognised sites of conservation importance

15.2.2.1 All recognised sites of conservation importance, including Tai Lam Country Park and the Conservation Area have been avoided and will not be directly encroached by any developments under the Project.

15.2.3 Avoidance of Direct Impact on Watercourses of Higher Ecological Value

15.2.3.1 As discussed in **Section 1.7.10**, three streams with endemic crab species including Stream Catchment 7 and Stream Catchment 6 (~1.6km in total) in Area 1 and Stream Catchment 4 (~475m) in Area 3 of the PDA have records of the crab *S. zanklon*, which is listed as being of global concern and is considered to be endangered by IUCN. In order to protect these habitats, the current design has retained with 15m buffer on both sides of these watercourses. In particular, the watercourse in Catchment 7 will be extended to connect with the revitalised Yuen Long Nullah. In addition, the Yeung Ka Tsuen Ecologically Important Stream (EIS) to the south of YLS development has also been totally avoided for conservation purpose.

15.2.4 Avoidance of impacts to Tai Tong (Pak Sha Tsuen) Egretry

15.2.4.1 As discussed in **Section 8.6.3**, there is no egretry within the PDA. The Tai Tong (Pak Sha Tsuen) Egretry is located to the west of Kung Um Road near Area 2 and is the only egretry in the vicinity which supports a low number of breeding birds. Following the results of flightline surveys conducted under this Study, the RODP has been adjusted to avoid encroachment on this egretry and impacts to major flightlines by providing Non-building Areas (NBAs) to nearby residential zones. In view of the long separation distance, given the small size of this egretry, the proposed zoning and NBA will not result in significant ecological impacts (including both disturbance or displacement from their flight paths in the foraging area) to the breeding egrets.

15.2.5 Avoidance of Encroachment on Existing Villages

15.2.5.1 A number of existing villages lie within and immediate surrounds of the PDA. These include villages in Tong Yan San Tsuen (TYST), Shan Ha Tsuen, Lam Hau Tsuen, Tin Liu Tsuen, Sham Chung Tsuen, Muk Kiu Tau Tsuen and Pak Sha Tsuen. During the planning stage, encroachment on these existing villages have already been avoided as much as practicable except TYST, where majority of the village houses have zoned as “Residential – Zone 5 (Existing Development Area)” (“R5 (EDA)”). Moreover, a Local Open Space (“LO”), with a width at least 20m, has been planned for the better integration of between the high rise buildings and the existing villages. For other existing residential developments and village houses with in the PDA, it is proposed to preserve these developments under this Project.

15.2.5.2 Three “R6” zone have been reserved in each of the three village rezoning sites, adjacent to the existing indigenous villages for the re-provisioning of the affected village houses (Sites VR1.1, VR2.1, and VR3.1), thus the relocated village houses would be in the vicinity to the existing communities.

15.2.6 Avoidance of Encroachment on Built Heritages

15.2.6.1 The existing structures of cultural and historical importance within and in the vicinity of the PDA are actively avoided through careful planning. These include the declared monument of Cheung Ancestral Hall in Shan Ha Tsuen, Grade 3 historic building of Yeung Hau Temple and the main temple premises of Kam Lan Monastery. The Cheung Ancestral Hall is located outside the PDA boundary, and both of the temples are designated as "Institution or Community" ("IC") under the RODP.

15.3 Environmental Designs Recommended to Minimise and Mitigate Environmental Impacts

15.3.1.1 Other than initiatives to avoid environmental impacts as summarised in **Section 15.2**, efforts have been exercised to minimise those inevitable impacts. The need for any environmental designs required to mitigate the associated impacts have also be identified and will be implemented as appropriate. A summary of these approaches is given below.

- Adopting stepped building heights;
- Providing greening for YLS;
- Catering for the future generation of EFTS extended from HSK NDA;
- Clearance of existing odour sources;
- Reducing commuting traffic by providing local employment opportunities;
- Encouraging green commuting within and beyond Yuen Long South;
- Screening of OU buildings;
- Covered PTIs;
- Providing direct noise mitigation measures;
- Design of noise barrier to avoid bird collision;
- Providing sustainable drainage system within PDA;
- Providing sewer system within the PDA and sewage treatment works (STW);
- Minimising encroachment on agricultural lands and woodland; and
- Implementation of environmental monitoring and auditing system during construction phase.

15.3.2 Adopting Stepped Building Heights

15.3.2.1 As discussed in **Section 3.4.2**, through the tapering of development intensities and building heights from north to south, YLS will integrate with its surrounding settings by higher density developments in the north near Yuen Long New Town (with approximate building height of 120mPD) and transitioning to medium and low-density development in the south (with approximate building height of 40-60mPD) to blend in with the rural setting and Tai Lam Country Park. Low-density developments are planned in the TYST area to cope with the existing low-density and low-rise residential developments. Furthermore, development intensity of the

gateway area of TYST would be higher with PR of 4. As existing buildings in TYST are low-rise in nature, lower PR of 1 and 1.5 are proposed to complement the existing residential communities. The benefits of stepped buildings height from Yuen Long New Town towards the Tai Lam Country Park are to avoid and minimise "wall effect" developments along the boundary of Country Park as far as possible.

15.3.3 Providing Greening for YLS

15.3.3.1 As explained in **Section 1.7.10**, greening is also another key consideration in developing the RODP. According to the current proposal, more than 50ha of land (including Open Space, Green Belt, Amenity area and Agricultural zone) are reserved for retaining the existing landscape characteristic and providing new green landscape throughout the PDA, which aim to enhance the local character, green landscape and linkage between urban, rural and nature areas.

15.3.3.2 The proposed enhancement scheme comprises new planting space to visually soften the surrounding environment, mainly planting in Open Space & Primary Amenity Areas, Greenbelt & Secondary Amenity Areas, Residential Areas, Roadside & Major Corridors and Scenic Routes. Thematic planting has also been considered in the RODP as part of the enhancement measures for both landscape and visual aspects. Species proposed for YLS are selected according to their ecological function and their attractive foliage, lush colours, and varied shapes in order to create an impressive visual amenity.

15.3.4 Catering for the Future Generation of EFTS extended from HSK NDA

15.3.4.1 The RODP has reserved a corridor for the extension of the EFTS proposed in HSK NDA, with possible connection to West Rail Tin Shui Wai Station. The corridor navigates through the storage/workshops, residential, commercial and other land reserves within the PDA. This EFTS would encourage both existing and future communities to make use of this environmentally friendly transportation travelling to other urban areas outside YLS, instead of road based transportation. This would help reduce subsequent vehicular emissions and road traffic noise impacts within and in the vicinity of YLS.

15.3.5 Clearance of Existing Odour Sources

15.3.5.1 Currently, there are six livestock farms in the YLS PDA, including three pig farms and three chicken farms. For the sake of minimising the risk of avian influenza in Hong Kong, the Government has since 2008 adopted a policy of containing the scale of live poultry trade in Hong Kong, with the total rearing capacity of live chickens capped at 1.3 million. Under the existing statutory provisions, there are constraints in relocating the affected chicken farms due to government policies.

15.3.5.2 Three livestock farms (one pig farm and two chicken farms) that are located in the central locations of the PDA have inevitably to be cleared.

15.3.5.3 For the chicken farms, there is no change to the statutory restriction on chicken farm relocation at this stage. Therefore, the chicken farms could not be relocated. In order to strike a balance among public opinions, minimise anticipated interface issue with existing and future residents and to minimise odour impact, only one chicken farm

located at the southern fringe of the PDA is planned to be retained. Sufficient buffer area has been reserved in the RODP.

15.3.5.4 For the pig farms, taken into account the adverse odour impact and the opinions of the public, the two pig farms at the southern end of Kung Um Road could not be retained under the RODP. The areas will be rezoned as "G" for government depot and serve as part of the buffer between the retained chicken farm and sensitive land uses.

15.3.6 Reducing Commuting Traffic by Providing Local Employment Opportunities

15.3.6.1 Currently, a large area of YLS PDA of approximately 100ha is occupied by brownfield operations such as warehouses, open storage yards, vehicle repairs, waste recycling yards and other rural industrial workshops. While it is recognised that the industrial uses provide backup services to the logistics industry and employment opportunities for locals, the intermingled land uses have been exerting various issues and nuisance on the environment, such as the noise nuisances from truck movements, odour and visual nuisance from operational activities, etc.

15.3.6.2 As discussed in **Section 1.7.3**, taking advantage of the close connection with the highway network, the Employment Belt has been proactively proposed along YLH where the area is easily accessible to the highway and avoiding freight traffic travelling through the PDA. The RODP has proposed zonings for the northern part of Area 1 as "Storage Use", "Storage and Workshop", "Open Storage" and "Mixed Use". Approximately 1.4 ha of land for open storage use and approximately 10 ha of land for storage and workshop uses are proposed, which are intended to support industrial and logistic operations.

15.3.6.3 Multi-storey industrial sites are planned at the "Storage Use" and "Storage and Workshops" zones to consolidate the existing operations on brownfield. To further segregate industrial land uses and residential areas, a network of open spaces has also been planned while the setback distance has been allowed to address road traffic noise issues, the multi-storey buildings could also serve as a screen to further reduce the impacts of road traffic noise from YLH on the residential areas in the southern part of Area 1.

15.3.7 Encouraging Green Commuting within and beyond Yuen Long South

15.3.7.1 As discussed in **Section 1.7.12**, comprehensive networks of pedestrian walkways and cycle tracks are planned within the PDA to promote green commuting. Key pedestrian corridors will be provided with amenity planting to enhance the walking environment for pedestrians. Besides, cycle tracks of about 11km will also be built across the PDA and be connected to other districts such as Ping Shan, Tin Shui Wai New Town and Yuen Long New Town where practicable. There are two connection points between the proposed cycling network of the PDA and the external existing / planned cycle tracks. One of the connection points is along Fui Sha Wai South Road near TSWW Interchange, and the other one is along west of Long Tin Road near TYST Interchange. A cycle track and a walking trail are also proposed to connect the PDA with Tai Lam Country Park. All these measures are proposed to facilitate walking and cycling activities across the PDA and in the wider community.

15.3.7.2 Space is also reserved for a possible addition of EFTS which may be able to connect the PDA and HSK NDA. If the EFTS is materialised by the future relevant project proponents, this will give residents and visitors another option for green commuting.

15.3.7.3 In addition to promoting green commuting, it is also important to minimise commuting. Currently, there is a lack of community facilities within the PDA and residents would need to travel to Yuen Long New Town for their daily necessities. The RODP has planned G/IC facilities such as educational uses, sports and recreational facilities, elderly homes, clinic and community centres, etc. These facilities could reduce travel distance/trips of both existing and future residents of the near community. On the other hand, this can also relieve the stress on public transport systems and the associated environmental issues.

15.3.8 Screening of OU buildings

15.3.8.1 According to the current construction phasing programme (see **Section 3.7**), the construction of the multi-storey buildings in the Employment Belt in Area 1 would be completed by Stage Two in Year 2033 and prior to the occupation of the planned residential buildings in Area 1, which would be completed by Stage 4 in Year 2038. While the setback distance would be sufficient to abate road traffic noise impacts, the multi-storey buildings could also serve as a barrier to further reduce the impacts of road traffic noise and industrial noise from YLH and the existing industrial area on the residential areas in the southern part of Area 1.

15.3.9 Covered PTIs

15.3.9.1 Four PTIs are proposed under the current project, three are within the PDA and one is outside the PDA near Yoho Town in Yuen Long. All these PTIs will be covered to minimise the operational noise to existing and planned receivers.

15.3.10 Providing Direct Noise Mitigation Measures

15.3.10.1 A road traffic noise assessment (see **Section 5.6**) has been conducted for the existing and planned NSRs (including residential units, schools, etc.) within the 300m assessment area of the PDA and the associated infrastructure. The current planning has allowed setback distances as much as practicable. Besides, the current planning has located commercial building uses between internal roads in the PDA and residential uses so as to provide as much noise screening as possible. Nevertheless, some of the noise sensitive receivers would still require direct noise mitigation measures to comply with all the legislative requirements, including 1) absorptive vertical barriers and cantilevered noise barriers along some sections of Project Roads and Kung Um Road; 2) LNRS on some road sections; 3) semi-enclosures or full enclosures at primary distributor roads at TYST Interchange and Kung Um Road; 4) nullah features / barriers along some sections of Yuen Long Nullah (Kiu Hing Road); 5) provision of acoustic windows for some planned public and private housing and 6) alternative building orientation for some planned schools are proposed to alleviate adverse traffic noise impact on the affected NSRs . More detailed description of those direct noise mitigation measures are given in **Section 5.6**.

15.3.11 Design of Noise Barrier to Avoid Bird Collision

15.3.11.1 In order to minimise the potential impact of bird collision, only solid noise barrier will be used during construction phase. Transparent panels for noise barriers and

noise enclosures will be avoided as far as possible during operation phase. If transparent panels will be used, stickers or other relevant measures will be applied to increase the visibility of the panels to avoid bird collisions.

15.3.12 Providing Sustainable Drainage System within PDA

15.3.12.1 Flood retention facilities such as flood retention tank/ lake/ pond are proposed to cater for the additional runoff caused by the PDA. These storage devices retard residence time and alleviate peak flows. Instead of expelling flood waters, these measures hold up water and therefore can maintain steady river flows during drier periods.

15.3.12.2 As discussed in **Section 1.7.8**, in order to mitigate the increased flow from the development, a retention tank at Area 1 has been proposed to attenuate the peak flow, such that the peak outflow to the existing Yuen Long (West) Nullah would attain insignificant change in flow rate. The area and storage volume of the retention tank in Area 1 are about 2000m^2 and 11,760 m^3 respectively.

15.3.12.3 A retention lake and Hillside River Corridor have also been proposed at the upstream of Tin Tsuen channel (along Lam Tai East/West Road) to temporarily store the 50 year peak runoff from the existing hillside catchment and catchments from proposed development in Areas 2 and 3, and hence to mitigate the increase in peak surface runoff from the PDA. The retention lake and Hillside River Corridor will also serve for amenity purposes and enhance the landscape and visual environment. According to the DIA under the Study, the area and storage volume of the retention lake are approximately 0.6ha and 13,000 m^3 respectively. The Hillside River Corridor runs along the PDA boundary and is about 575m and 0.37ha, with a storage volume of about 3,200m^3.

15.3.12.4 In addition, a reedbed/retention pond (approximately 4ha) has been proposed at the south of Area 3 under the RODP. The reedbed can act as a sponge that traps and slowly releases surface water. Reeds and their root mats slow the speed of flood waters and distribute slowly over the area. In addition to the benefits of flood control, the permeability of reedbed allows infiltration of rain water that replenishes groundwater levels. This reedbed also serves as a visual element for the local community.

15.3.13 Providing Sewer System within the PDA and Sewage Treatment Works

15.3.13.1 There has been long-term improvement in Water Quality Objectives (WQO) compliance rate and Water Quality Index (WQI) grades of Hong Kong's inland water courses (see **Section 6** for water quality baseline conditions). Despite the trend of general improvement, according to the EPD River Water Quality in Hong Kong 2014, the water quality of the water courses including those in YLS forming Yuen Long Creek have a range of Water Quality Index grading from "Bad" to "Fair". Currently, there is no STW within the PDA, and the watercourses are still subject to discharges from various sources such as livestock farms, unsewered village houses and industrial establishments. The estimated existing unsewered population within the scattered villages and brownfield sites within the PDA is approximately 280 persons (about 100 households), and the estimated employment is about 4,500 persons within the existing brownfield sites within the PDA. With the proposed STW implemented under the Project, all the areas within the PDA boundary will be provided with sewers that convey the sewage to the new STW. Hence, all existing

unsewered population within the PDA will be sewered and the watercourses in the vicinity are anticipated to have potential benefits from the improvement in water quality.

15.3.13.2 The sewage flows and loads generation from the YLS PDA are estimated and the proposed sewage treatment and disposal strategy for YLS are accordingly formulated, which involve discharging the YLS sewage flows by the proposed sewerage system to a new on-site STW in the southern part of the PDA for treatment, followed by the reuse of reclaimed water as non-potable use such as toilet flushing within PDA and adjacent developments. Equivalent pollution loading from surplus reclaimed water in YLS STW (upon offsetting by credit gained from the existing livestock farms) could be discharged to Yuen Long Nullah adjacent to the STW with strict compliance to the requirement for "No Net Increase in Pollution Loading", while reedbed is provided a further polish to the surplus reclaimed water that discharge to Yuen Long Nullah and Deep Bay.

15.3.14 Minimising Encroachment on Agricultural Lands and Woodland

15.3.14.1 The RODP has preserved all woodland patches of significant sizes in the PDA and avoided encroachment and fragmentation of the mature woodland behind Shan Ha Tsuen through adoption of separate road junctions located away from the preserved woodlands and the largest agricultural land near Shan Ha Tsuen. The RODP will also retain three sections with buffers which have records of the endemic crab species (i.e. *S. zanklon*). Translocation of this species can be conducted to these three streams or the proposed 575m river corridor should they occur in those fragmented watercourses in the lowlands.

15.3.15 Implementation of Environmental Monitoring and Auditing System during construction phase

15.3.15.1 In addition to the mitigation measures as described above (see more details in the Project Implementation Schedule given in the Environmental Monitoring and Audit Manual), a comprehensive environmental monitoring and auditing programme would be implemented to cover various aspects of concern. An independent environmental checker would also be employed to ensure that all the necessary mitigation measures are implemented in a timely and orderly manner.

15.4 Summary of Key Environmental Problems Avoided and Sensitive Areas Protected

15.4.1.1 **Sections 15.2 and 15.3** have summarised the key approaches adopted in the RODP to avoid, minimise and mitigate environmental impacts. Some of these approaches have contributed to avoid a number of environmental problems and to protect a number of environmental sensitive areas. The following table presents the key environmental problems that have been avoided and any sensitive areas protected by these approaches.

Table 15.1 Key environmental problems avoided and sensitive areas protected

Design Approaches	Key Environmental Problems Avoided and Sensitive Areas Protected
Avoidance of recognised sites of conservation importance	• All the recognised sites of conservation importance, including Tai Lam Country Parks, Conservation Area have been avoided and will not be encroached by any developments under the Project.
Avoidance of direct impact on watercourses of higher ecological value	• The concerned sections are proposed to be retained with 15m buffer on both sides of the watercourses, in which one of the watercourses is proposed to be extended to connect with the revitalised Yuen Long Nullah.
Avoidance of impacts to Tai Tong (Pak Sha Tsuen) Egretry	• The RODP has been adjusted to avoid impacts to major flightlines and has also provided NBAs in Residential use.
Avoidance of encroachment on existing villages	• Preserve existing residential development and village houses.
Avoidance of encroachment on built heritages	• Monument and graded buildings have been avoided and the associated heritage resources are protected.
Adopting stepped building heights	• Reduce "wall effect" along hilled areas
Providing greening for YLS	• A Green Zone of about 15ha will be provided to enhance the local character, green landscape and linkage between urban, rural and nature areas.
Catering for the future generation of EFTS extended from HSK NDA	• The EFTS would encourage both existing and future communities to make use of environmentally friendly transportation travelling to other urban areas outside YLS, instead of road based transportation. This would help reduce subsequent vehicular emissions and road traffic noise impacts within and in the vicinity of YLS.
Clearance of existing odour sources	• Only one chicken farm could be retained with buffer area reserved on plan.
Reducing commuting traffic by providing local employment opportunities	• Taking advantage of the close connection with the highway network, an "Employment Belt" has been proactively proposed along YLH where the area is easily accessible to the highway and avoiding freight traffic travelling through the PDA. Moreover, it is anticipated that about 10,500 jobs will be created and these local job opportunities can help reducing need for community traffic and hence the associate air emissions and traffic noise nuisance for the existing sensitive receivers along the commuting routes.
Encouraging green commuting within and beyond Yuen Long South	• Comprehensive networks of pedestrian roads and cycling tracks are planned within the PDA to promote green commuting. On the other hand, this can also relieve the stress on public transport systems and the associated environmental issues.
Screening of OU buildings	• While the setback distance would be sufficient to abate road traffic noise impacts, the multi-storey buildings could also serve as a barrier to further reduce the impacts of road traffic noise from YLH on the residential areas in the southern part of Area 1.
Covered PTIs	• Operational fixed noise impacts from proposed PTIs to NSRs are avoided/minimised.

Design Approaches	Key Environmental Problems Avoided and Sensitive Areas Protected
Providing direct noise mitigation measures	• All noise sensitive receivers within the RODP are protected to achieve the statutory traffic noise criterion.
Design of noise barrier to avoid bird collision	• The potential impact of bird collision could be minimised.
Providing sustainable drainage system within PDA	• Direct release of surface runoff into Yuen Long Nullah could be avoided and the ecological sensitive receivers in the vicinity are protected as much as practicable.
Provision of Sewage Treatment Works and Treated Sewage Effluent Reuse	• All the areas within the PDA boundary will be provided with sewers that convey the sewage to the new sewage treatment plant. Watercourses within the vicinity are anticipated to have potential benefits from the improvement in water quality.
Minimising encroachment on agricultural lands and woodland	• The RODP has preserved all woodland patches of significant sizes in the PDA and avoided encroachment and fragmentation of the mature woodland behind Shan Ha Tsuen through adoption of separate road junctions located away from the preserved woodlands and the largest agricultural land near Shan Ha Tsuen.
Implementing an EM&A during construction phase	• Ensure that all the recommended measures are in place

15.5 Estimated Population Protected from Various Environmental Impacts

15.5.1.1 As mentioned in **Section 1.7**, a Sustainable Urban Drainage System (SUDS) has been introduced in YLS in which high flows will be directed into stormwater attenuation and retention lakes and tanks during large rainfall events to avoid flooding. The existing villages within PDA at YLS with a total population of approximately 2400.

15.5.1.2 In addition, with the implementation of proposed noise mitigation measures, all the total planned population assigned in YLS PDA (approximately 88,000) will be protected from road traffic noise.

15.6 Environmental Benefits of Environmental Protection Measures Recommended

15.6.1.1 Mitigation measures have been recommended to further reduce the environmental impacts during the construction and operational phases of the project. Key recommended mitigation measures and the associated benefits are summarised in **Table 15.2** below.

Table 15.2 Key recommended mitigation measures and associated benefits

EIA Aspects	Key recommended mitigation measures and associated benefits
Air Quality	• Implementing relevant control measures as required in the Air Pollution Control (Construction Dust) Regulation to minimise dust generation; • Watering once per hour on exposed worksites and haul road; • Potential odour sources associated with the proposed sewage pumping station should be enclosed; • Negative pressure should be maintained within the proposed sewage pumping station;

EIA Aspects	Key recommended mitigation measures and associated benefits
	• Installation of deodouriser with an odour removal efficiency of at least 95% to control odour emission from via ventilation exhaust of the proposed sewage pumping station; • Exhaust of the deodouriser at the proposed sewage pumping station should be oriented away from sensitive receivers and vertically upwards to avoid direct facing to any sensitive receivers; and • Maintenance of deodouriser at the proposed sewage pumping station should be regularly conducted to ensure good condition.
Noise	• Selection of quiet plant and use of movable noise barriers / acoustic shed / noise insulating fabric during construction phase to reduce construction noise impacts to representative NSRs • Provision of direct noise mitigation measures to reduce traffic noise impacts on representative NSRs
Water Quality	• Implementing the mitigation measures recommended for control of suspended solids release • Implementing site practices as outlined in ProPECC Note PN 1/94 to control and minimise site runoff and drainage • Install and maintain roadside gullies and oil / grease interceptors for removal of pollutants from storm water
Sewerage and Sewage Treatment	• Propose new village sewerage system for the unsewered villages within PDA to ensure proper handling of sewage flow • Rese of reclaimed water (i.e. flushing) within the PDA and adjacent developments
Waste Management	• Good waste management and control practices to avoid generation of excessive amount of waste materials • Good site practices to alleviate impacts related to waste management such as dust, odour and noise • Implementation of a trip-ticket system promulgated under DEVB TC(W) No. 6/2010 to monitor off-site delivery of surplus inert C&D materials • Handling of chemical wastes in accordance with the Code of Practice on the Packaging, Labelling and Storage of Chemical Wastes, and disposal of chemical wastes at licensed chemical waste recycling/ treatment facilities • Employ reputable licensed waste collectors for disposal of general refuse
Land Contamination	• Carry out suitable remediation measures according to the results from future Site Investigation prior to construction after land resumption
Ecology	• An off-site/ near-site for compensatory woodland planting (~12ha) has been proposed to provide at least 2.4ha of compensation woodland, i.e. a compensation ratio of at least 1:1 to adequately mitigate the impacts from woodland loss • Retain and/or transplant the plant species of conservation importance affected as far as possible • Three lowland watercourses are proposed to be retained (and enhanced/ reinstated where possible) to increase the ecological value of the PDA and to help maintain species diversity through reduction of habitat fragmentation • Zoning of these watercourses and associated buffers as 'Green Belt' will aid to preserve the integrity of the good quality watercourses and their riparian corridor which are largely natural/semi-natural with good water quality and have high ecological value providing linkages and corridors. • Preservation of habitats with ecological values • Implementing good site practices

EIA Aspects	Key recommended mitigation measures and associated benefits
Fisheries	• Implementing good site practices
Landscape and Visual	• Sensitive landscape design of land formation edge, as well as buildings and structures • Provide temporary landscape on temporary construction • Implement greening measures, including thematic planting, vertical greening, green roofs, road verge planting and peripheral screen planting where practical to improve the landscape and visual aesthetics • Compensatory Tree Planting for all felled trees to replace affected trees • Woodland enhancement by planting of native tree species and transplantation of affected trees • Sensitive design of streetscapes (e.g. paving, signage, street furniture, lighting etc.) and viaduct structures to minimise visual intrusion
Cultural Heritage	• Declared monument and graded buildings avoided.

15.7 Environmental Enhancements

15.7.1 General

15.7.1.1 Other than the approaches to avoid, minimise and mitigate environmental impacts as summarised in **Section 15.2 and Section 15.3**, a series of environmental enhancement works has also been proactively considered by the Project Proponent, as explained in the following sections.

15.7.2 Revitalisation of Yuen Long Nullah

15.7.2.1 Yuen Long Nullah is a key element in the YLS, receiving water flow from hillside adjacent to Tai Lam Country Park in the south, and connecting to water channel in Yuen Long New Town Centre in the north. The nullah is currently channelized and lined with concrete (approximately 2.5km long), with narrow traffic roads on both sides (i.e. Kung Um Road and Kiu Hing Road) and limited area for pedestrian passage/amenity.

15.7.2.2 As part of this Project, major rehabilitation works without decking are proposed for the 0.7km southern section of the Yuen Long Nullah. A provision of approximately 4m-wide footpath and cycle track which will be accompanied by amenity plants and landscaped open space. The vegetation will introduce a green spine and improve the visual appeal of the watercourse. The revitalised nullah with soft and hard landscaping will be promoted as a place for leisure for both existing and future residents as major visual corridor for the community and help improve the micro-climate and contributing to alleviate certain urban heat island effect. Partial decking would be used to provide sufficient space along the upper section of Yuen Long Nullah.

15.7.2.3 Benefited from the amenity plants and landscape open space, the revitalised Yuen Long Nullah will also serve as a major visual corridor in YLS linking the PDA and the surrounding natural landscape assets including those in Tai Lam Country Park and the Conservation Area. The vegetation along the nullah would also improve the micro-climate and contribute to alleviate certain urban heat island effect.

15.7.3 Rezoning of Degraded Brownfield Sites

15.7.3.1 Currently, approximately 100ha of land within the PDA is degraded brownfield sites used for industrial purpose and the original agricultural lands have already been disturbed by brownfield operations. The brownfield area is mainly zoned as "Undetermined" and the remaining ones are zoned as "I(D)", R(D) etc. on the approved Outline Zoning Plan TYST OZP No. S/YL-TYST/10.

15.7.3.2 The original agricultural lands have already been disturbed by brown field operations. Various ecological habitats have been identified within the PDA and its vicinity (see **Chapter 8** for details). They include an active egretry in Tai Tong, woodland patches and a large piece of agricultural land near Tong Yan San Tsuen, and few sections of watercourses of ecological value within the PDA. The RODP has considered the value of these important environment receivers and has proposed to conserve them.

15.7.3.3 During the earlier planning stage, the boundaries of the PDA are designed to avoid encroachment onto these ecologically important resources as far as practicable, such as the EIS near Yeung Ka Tsuen, which is now located outside the PDA boundary. Moreover, the planned roads and road junctions have avoided the woodlands and the agricultural land as far as practicable, to avoid and minimise disturbance and fragmentation of those habitats.

15.7.3.4 Therefore, the Study has proposed rezoning of these areas, so that the ecologically important resources within these "Undetermined" landuses can be conserved. Nevertheless, these would still be approximately 6.5ha of woodland of various ecological quality that would be inevitably affected, and hence would require compensation (see **Chapter 8** for more details).

15.7.3.5 On this basis, landuse of the areas within the PDA near these ecologically important resources have been proactively designed to minimise human disturbance and provide additional setback where practicable. For instance, zonings of district open space (DO), local open space (LO) and amenity areas (A) are proposed at the southern tip of Area 3 which is near Tai Lam Country Park, Yeung Ka Tsuen EIS and Tai Tong Egretry. These land uses are characterized by less human disturbance, and offer additional setback to the ecological habitats.

15.8 Compensation Areas

15.8.1 Woodland Compensation

15.8.1.1 As discussed in **Chapter 8**, only the impact resulting from the cumulative loss of hillside secondary woodlands (~2.42ha in total) is considered of moderate significance and mitigation is considered as necessary. Given the limited space within the RODP, on site woodland compensation is considered not practicable. Loss of hillside secondary woodlands is to be compensated by woodland planting in the adjacent hillside area to the west of the PDA. The proposed woodland compensation site is approximately 12ha in the hillside area currently zoned as "Conservation Area". A minimum of 2.4ha of compensation woodland should be provided within the proposed planting site (~12ha), i.e. a compensation ratio of at least 1:1 should be provided. The proposed planting site is not uniformly sloping. It includes some steeper areas with large boulders and some areas along the ridges are more gently sloping (slope gradient <35˚) that are considered suitable for tree planting. The proposed site has been sparsely planted with exotic plantation species

of *Pinus elliottii*. The general ground area is sparsely covered with shrubby grassland species such as the fern species *Dicranopteris pedata* and the shrub species *Baeckea frutescens*. The site is considered to be moderately eroded and exposed. Plant species should be selected to include a mix of species with pioneering characteristics (fast-growing/ light-tolerant/ drought-tolerant/ wind-tolerant, etc.) and native species. The planting design should avoid intense competition between native species and exotic species otherwise the former would be out-competed by the latter. In general, fast-growing exotic species such *Melaleuca cajuputi subsp. cumingiana* and *Acacia sp.* can be planted along the most severely eroded ridges and wind-exposed areas. Native species should be planted in groups instead of inter-planting between exotic species to minimise competition between exotic and native species.

15.8.2 Tree Compensation

15.8.2.1 As discussed in **Chapter 11**, there will be a certain amount of vegetation clearance and trees would need to be felled. Based on the broad brush tree survey results, approximately 63% of exiting trees within the development boundary will be potentially affected.

15.8.2.2 There are 3 Old and Valuable Trees (OVTs) found in the assessment area, but outside the PDA; however there are 77 identified Important Trees (including POVT and rare/ protected species) as per LAO PN No. 7/2007. It is anticipated that 34 of them are in direct conflict with the proposed development. A detailed Tree Felling Application process will be carried out at a later detailed design stage, to finalise tree treatment and allocate compensatory planting areas including available open space, parks and streetscape. Some compensation planting and transplanting of trees will be required only for mitigating landscape and visual impacts. Woodland mix species whips will be proposed along the hillsides for the off-site woodland compensation.

15.8.2.3 Compensatory tree planting for felled trees shall be provided to the satisfaction of relevant Government departments. Required numbers and locations of compensatory trees shall be determined and agreed separately with Government during the Tree Removal Application process under DEVB TCW No. 10/2015. Compensatory planting is proposed at the potential open areas such as open spaces, amenity areas, open areas of the streetscapes, as well as the open areas within development lots.